

Mohamed Salah vs Kylian Mbappé: An Attacking Performance Analysis (2017/18–2024/25)

Executive Summary

This project analyzes Mohamed Salah and Kylian Mbappé's attacking performances from the 2017/18 to 2024/25 seasons to determine who was the more efficient and consistent attacking contributor. Using SQL in Google BigQuery, player performance data was analyzed across offensive output, finishing efficiency, chance creation, and season-to-season consistency.

The analysis found that Kylian Mbappé produced higher average attacking output, particularly in goals per 90, goal contributions, expected attacking contributions, and creative actions. However, Mohamed Salah demonstrated greater consistency, recording lower variation across every attacking metric measured through standard deviation.

The findings suggest that while Mbappé produced a higher attacking peak, Salah maintained a more stable level of elite offensive performance across the eight-season period.

Research Question

Between Mohamed Salah and Kylian Mbappé, who was the more efficient and consistent attacking contributor from 2017/18–2024/25?

Data Sources

- FIFA player database
- FBref player performance statistics
- Seasons analyzed: 2017/18–2024/25

- Database platform: Google BigQuery
- Query language: SQL

The dataset combined player attributes and season statistics to evaluate attacking performance using per-90 metrics rather than only total statistics.

Methodology

The analysis followed these steps:

1. Imported and explored the football dataset in Google BigQuery.
2. Verified player identities and available seasons.
3. Identified duplicate player records.
4. Standardized player names to ensure accurate comparisons.
5. Selected attacking metrics representing:
 - a. Offensive production
 - b. Finishing efficiency
 - c. Chance creation
 - d. Consistency
6. Calculated career averages across the 2017/18–2024/25 period.
7. Compared season-by-season performance.
8. Measured consistency using standard deviation.

Data Cleaning

During the data validation process, Mohamed Salah appeared under two different long_name values:

- Mohamed Salah Ghaly
- Mohamed Salah Hamed Ghaly

Both records represented the same player. The entries were combined into one category:

Mohamed Salah

This ensured all eight seasons from 2017/18–2024/25 were included in the analysis.

Comparison Categories

1. Offensive Output

Metrics selected:

- Goals per 90 (Per 90 Minutes_Gls)
- Assists per 90 (Per 90 Minutes_Ast)
- Expected attacking contribution per 90 (Per 90 Minutes_xG+xAG)

2. Finishing Efficiency

Metrics selected:

- Shot-on-target percentage (Standard_SoT%)
- Goals per shot (Standard_G_Sh)
- Goals per shot on target (Standard_G_SoT)

3. Chance Creation

Metrics selected:

- Expected assists per 90 (Per 90 Minutes_xAG / Expected_xA)
- Shot-creating actions per 90 (SCA)
- Goal-creating actions per 90 (GCA)

4. Consistency

Metrics selected:

- Season-by-season performance
- Average metrics across 2017/18–2024/25
- Standard deviation across attacking metrics

These metrics were selected to evaluate overall attacking influence rather than focusing only on goals and assists.

Average Attacking Comparison (2017/18–2024/25)

Metric	Mohamed Salah	Kylian Mbappé
Goals per 90	0.69	0.97
Assists per 90	0.32	0.28
Goal Contributions per 90	1.01	1.25
Expected Contributions (xG+xAG) per 90	0.93	1.13
Shot-Creating Actions per 90	4.43	4.64
Goal-Creating Actions per 90	0.68	0.81

Findings

The average attacking metrics indicate that Kylian Mbappé produced the higher overall attacking output.

Mbappé recorded higher values in:

- Goals per 90
- Goal contributions per 90
- Expected attacking contributions
- Shot-creating actions
- Goal-creating actions

Salah recorded a slightly higher assist rate, suggesting a stronger contribution through direct assists.

Consistency Analysis: Standard Deviation

Standard deviation was calculated across six attacking metrics to measure season-to-season variation.

A lower standard deviation indicates a more consistent performance level.

Metric	More Consistent Player	Standard Deviation
Goals/90	Mohamed Salah	0.147 vs 0.213
Assists/90	Mohamed Salah	0.104 vs 0.124
Goal Contributions/90	Mohamed Salah	0.201 vs 0.216
Expected Contributions/90	Mohamed Salah	0.154 vs 0.174
Shot-Creating Actions	Mohamed Salah	0.421 vs 0.580
Goal-Creating Actions	Mohamed Salah	0.183 vs 0.227

Findings

Mohamed Salah recorded a lower standard deviation than Kylian Mbappé in every metric analyzed.

This indicates that Salah maintained a more consistent attacking level throughout the 2017/18–2024/25 period, while Mbappé experienced greater season-to-season variation despite producing higher average output.

Conclusion

The analysis shows that Kylian Mbappé was the more productive attacking player based on average output. He recorded superior numbers in goals, goal contributions, expected contributions, and creative actions per 90.

However, Mohamed Salah demonstrated greater consistency across all measured attacking categories. His lower standard deviation indicates that he maintained a more stable level of performance throughout the eight-season period.

Overall, the results suggest that Mbappé produced the higher attacking peak, while Salah provided the more consistent attacking contribution over time.

Limitations

This analysis has several limitations:

- Only league performance was evaluated.
- Team tactics, teammates, and league strength were not adjusted for.
- Defensive contributions were excluded.
- Per-90 metrics measure efficiency but do not fully capture total workload or availability.
- The analysis focuses only on attacking contribution.

Project Skills Demonstrated

- SQL querying in Google BigQuery
- Data cleaning and validation
- Player identification and record standardization
- Statistical analysis using averages and standard deviation
- Sports performance analytics
- Data interpretation and communication